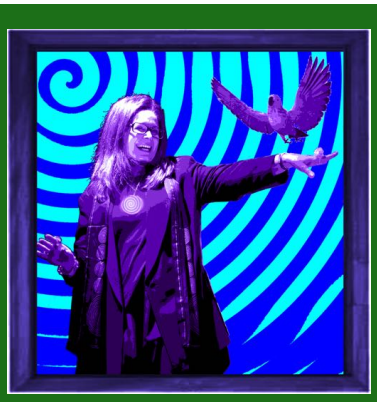
Alfred Jarry's Semantics : a web of disambiguation, mirrors, energy, machines, and potential collisions

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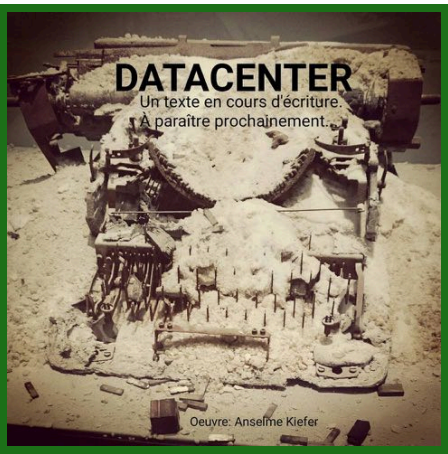
***Abstract*—This paper was given at www2016, Montreal, Canada, April 2016**

Keywords—web; disambiguation; pataphysics; energy; machines



IMAGINATION WWW'P

Alfred Jarry, in his brief thirty-four year lifetime that ended in 1907, laid the foundation for the avant-garde of the twentieth and twenty-first centuries' Western culture.[1] An innovator and rule-breaker, Jarry envisioned and formulated conscious 'Pataphysics, that is, the Science of imaginary solutions, the Science of the supplementary and virtual universe that investigates the particular and the dimension where art and science are on the same side of the coin.



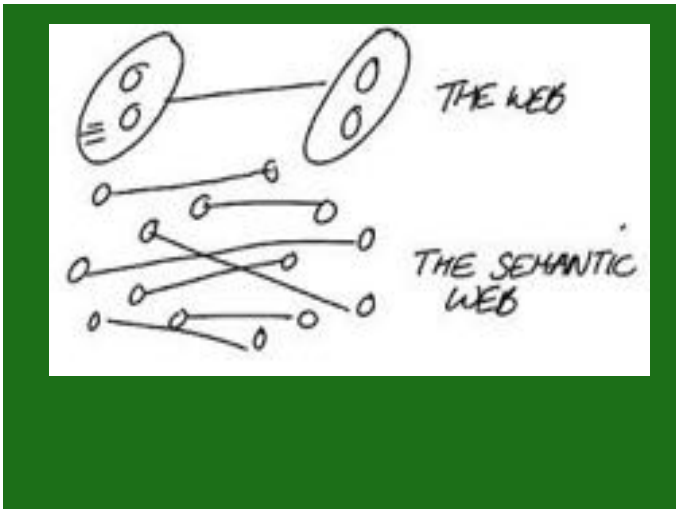
DATACENTER is a book by Léonel Houssam

In order to unpack the links and the leaps that connect Jarry's iconoclasm and semioclasm, at the time when typewriters were becoming more widely used, to our era of bits and bytes of information traveling from similar keyboards in a digital age, Jarry's definition of 'Pataphysics is ground zero.

Defined by Jarry most rigorously and painstakingly in *Exploits & Opinions of Doctor Faustroll, Pataphysician, neo scientific novel*, [2] he begins his definition of 'Pataphysics by specifying that "An epiphenomenon is that which is superinduced upon a phenomenon…" and he continues with the ground-breaking assertion that it "is the science of that which is superinduced upon metaphysics, whether within or beyond the latter's limitations, extending as far beyond metaphysics as the latter extends beyond physics. Ex: an epiphenomenon being often accidental, pataphysics will be, above all, the science of the particular, despite the common opinion that the only science is that of the general. Pataphysics will examine the laws governing exceptions, and will explain the universe supplementary to this one; or, less ambitiously, will describe a universe which can be—and perhaps should be—envisaged in the place of the traditional one, since the laws that are supposed to have been discovered in the traditional universe are also correlations of exceptions, albeit more frequent ones, but in any case accidental data which, reduced to the status of unexceptional exceptions, possess no longer even the virtue of originality." Then we read an explicit and succinct "DEFINITION. *Pataphysics is the science of imaginary solutions which symbolically attributes the properties of objects, described by their virtuality, to their lineaments.*" He goes further: "Contemporary science is founded upon the principle of induction: most people have seen a certain phenomenon precede or follow some other phenomenon most often, and conclude therefrom that it will ever be thus. Apart from other considerations, this is true only in the majority of cases, depends upon the point of view, and is codified only for convenience—if that! Instead of formulating the law of the fall of a body toward a center, how far more apposite would be the law of the ascension of a vacuum toward a periphery, a vacuum being considered a unit of non-density, a hypothesis far less arbitrary than the choice of a concrete unit of positive density such as water? (…) Universal assent is already a quite miraculous and incomprehensible prejudice. Why should anyone claim that the shape of a watch is round—a manifestly false proposition—since it appears in profile as a narrow rectangular construction, elliptic on three sides; and why the

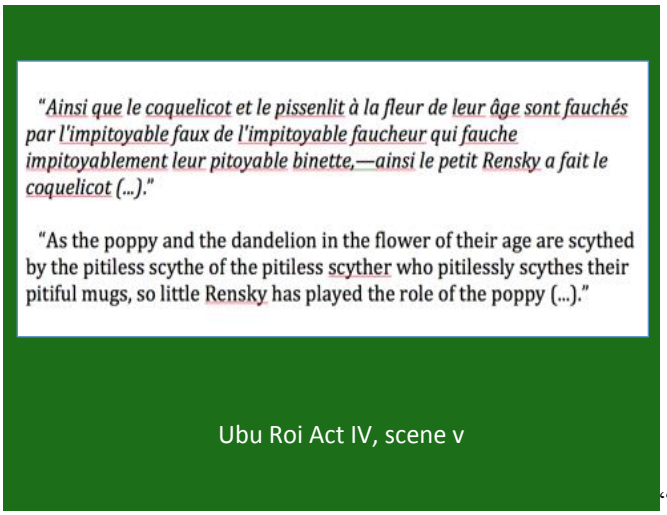
devil should one only have noticed its shape at the moment of looking at the time? —Perhaps under the pretext of utility. But a child who draws a watch as a circle will also draw a house as a square, as a façade, without any justification, of course; because, except perhaps in the country, he will rarely see an isolated building, and even in a street the façades have the appearance of very oblique trapezoids. We must, in fact, inevitably admit that the common herd (…) is too dimwitted to comprehend elliptic equations, and that its members are at one in a so-called universal assent because they are capable of perceiving only those curves having a single focal point, since it is easier to coincide with one point rather than with two. These people communicate and achieve equilib-rium by the outer edge of their bellies, tangentially. But even the common herd has learned that the real universe is composed of ellipses, and tradesmen keep their wine in barrels rather than cylinders."[3]

'Pataphysics does not replace but goes spiraling beyond metaphysics just as the Semantic Web is an extension springing from the current Web and is not its replacement.



For example, as Pablo Lopez suggests, because string theory is speculation based on theories that are or have been themselves speculative, such as general relativity and quantum mechanics, string theory cannot in fact be considered physics, but rather, should be understood as pataphysics. Likewise, string theory and quantum calculations do not portray an actual reality, but are mathematical pataphors.[4] Extended metaphors that create their own context, pataphors inform the patadata of the Semantic Web. As coined by Jim Hendler in 2008, "patadata" are "data interpreted through the truth of contradictions and exceptions" which is "as far from metadata as metadata extends from a databased representation of reality."[5] The Semantic Web, like all things "pata", foregrounds uncertainty, exceptions, and contradictions.[6] It supports what I call a messy, joyful yet sober, oscillating ontology. These language-based activities often seem to attribute equal consistency and equal weight and truthfulness to the, in fact, serendipitous semantics and structures of a data field. This furnishes ways to share and repurpose information thanks to the combinatory formats that become possible. As we learn from Jarry, who invented a code and culture of rotating points of view and kaleidoscopic perspectives, of new structures that accord equivalence to motifs and language observed and evaluated from different virtual angles, paradox as a semantic strategy creates parallel and potential universes. In one universe, we can say that we all generate data such as spending and income or storing wine in barrels or cylinders; let's say that, then, a government agency collects and links that data into one database of content-rich metadata. This is another, superinduced, universe. Likewise, the data about that data, patadata, with all of its equivocal and equivalent information, provide an accessible way to find supplementary input about any group.

In the 1890s, Jarry perfected this purposefully accidental, happily coincidental semantic tactic. He uses sampling and layering to showcase his innovative primary concepts of Pataphysics. In the benchmark tragi-comedy *Ubu Roi*,[7] first staged in 1896, Ubu concocts a very data-abundant epiphenomenal soliloquy of logical nonsense. At first glance this does not seem surprising. After all, the character Ubu and the science of Pataphysics developed among Jarry and his highschool friends at the Lycée de Rennes during the previous decade. Ubu took form from a much ridiculed physics teacher, Monsieur Hébert, whose name evolved to the silly-sounding *Ébé* and into the thunderous monstrosity *Ubu*. The pranksters performed puppet plays, inventing a series of misadventures for their anti-hero. Nevertheless, under 23 year-old Alfred Jarry's pen, Ubu's soliloquy gives ample evidence of a radical and visionary pataphysical esthetic. The deceptively simple speech integrates fragmentation, juxtaposition, shifting semiotic borders, creative destruction, playfulness, decentralization, mirroring and data linking. War and death and gardening merge to become a platform for an unexpected but meaningfully serendipitous semantic relationship. A pretentious and hilarious funeral oration by the greedy, blustery, cruel, and cowardly dictator King Ubu in *Ubu Roi* upends language itself while incorporating the coextensive aspects of 'Pataphysics and the semantic web:



"*Ainsi que le coquelicot et le pissenlit à la fleur de leur âge sont fauchés par l'impitoyable faux de l'impitoyable faucheur qui fauche impitoyablement leur pitoyable binette,—ainsi le petit Rensky a fait le coquelicot (...)*."[8] It is important to hear and see the original French text not only because it sounds so funny, which it does, but because even the magisterial translation by the pre-eminent Jarry translator Barbara Wright does not point us to connect all the dots of Jarry's opus that are contained in this apparently nonsensical tribute, and, by scaling up, to the textual machine of 'Pataphysics. The translation reads: "As the poppy and the dandelion in the flower of their age are scythed by the pitiless scythe of the pitiless scyther who pitilessly scythes their pitiful mugs, so little Rensky has played the role of the poppy (...)."[9]

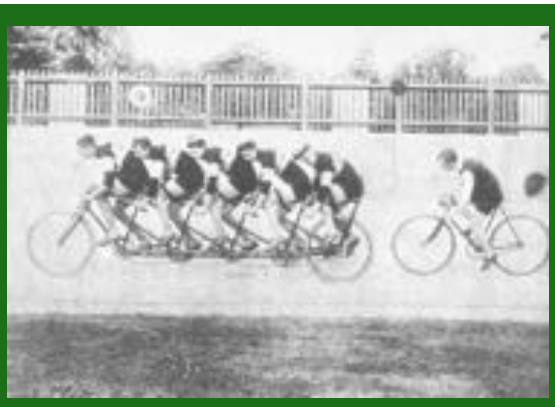
In miniature, this passage mimics Jarry's theories that privilege connectivity and creativity. It is a revelation of his consistent, foundational, poetics that articulate and organize spaces and shifts between universes beginning with his way of perceiving an object be it linguistic, semiotic, material, or virtual. Jarry's texts are decipherable by rendering transparent his machinery tightly regulating the tangential, the parallel, and the marginal.[10] This *Ubu Roi* quotation serves as a prototype, as Jarry's Beta version, of languages and tools for annotation by a Semantic Web. It is a self-contained system but it has unlimited creative imaginary potential. The sign series "*sont fauchés*, *faux*, *faucheur*, *fauche*" forms a network of signifiers and signified connected by letters, phonemes, morphemes, and so forth, as well as a signifying network that links this enunciation and this play to other texts that superficially have no correlation to these words operating in Act IV scene v of *Ubu Roi*. The material object *faux* indicates a scythe but is also the word that means "false" as compared to "true," setting in motion the links to the pataphysical theory of reciprocity, all things being equivalent, the plus and minus of the rotating *bâton-à-physique* or "physic stick" that appears in the *Ubu* plays, but also in others of Jarry's works that privilege the esoteric over the *ubuesque*. In addition, *faux* signifies a "fake" or a "counterfeit" and links this text to another pataphysical parameter: mirroring. *Être fauché* can mean to be cut by a scythe as a fact, a metaphor, or a pataphor. But *être fauché* also means "to be broke" as Jarry often was but Ubu was determined never to be. These opposites are themselves equal. *Faucher*, in familiar slang, also means to steal or "pinch" money. The semantic field shuttles from scythe to falsehood to mathematics to grass cutting to being mowed down in the street to questions of solvency and collection, and also confirms the malleability of language. The word *binette* is translated by Barbara Wright as "mugs," and while it does refer to a "face," in this case that of poor Rensky, in French it has a double meaning and *binette* is also a gardening tool, a hoe, similar in some respects to a scythe. The scythe links to yet another semantic search in another of Jarry's texts, *La Dragonne*.[11] Here it enters another network of the lance or musket or saber or dragoon, so-named because the guns the soldiers carried "breathed fire" like a dragon. The dragon, here also the "she-dragoon," (*la Dragonne*) routes us to a related domain of dragons, monsters, Jarry's definition of beauty,[12] as well as sexual ambiguity, hermaphrodites, and phallic swords subsumed in the category of Ubu's weapon of financial torture, itself thus linked to the network of the *faucher* metaphors and pataphors, that is, the *sabre à finance*, the finance saber, used to extract payment from Ubu's subjects. In Jarry's novel *La Dragonne*, the character *Fleur-de-sabre* loops back to the gardening data in King Ubu's tribute to his dead soldier. Her name itself links flowers and blades. Moreover, Jarry himself was an avid and competent fencer, photographed and painted with his saber. The semantic boundaries of art and life are mirrored by Jarry's textual universe. In a game of copies and counterfeits, the scythe also links to the saw, in French the *scie*, that Jarry makes clear is the semantic and phonemic building block of The Science, *La Science*, 'Pataphysics. We read in his 1902 *The Supermale, modern novel*,



"The SCIENCE with a big SCIE...".[13] This observation regarding The Science, made by the Supermale's sexual partner addressing him as the "Indian", the name of his sex-machine superman persona, again extends the semantic domain of "scythe" to the phallus. The search and the engine do not stop here. Alfred Jarry had no interest in

disambiguation; he designed a multiverse that integrates the contradictory and the paradox, antinomy and simultaneity. It transmits all potential meanings and allusions at once, and sets in alignment unexpected links in a semantic pattern. Patadata concerning the scythe, that is, information about its potential semantic equivalence to an imaginary universe, conflate death, as indicated in Ubu's parodic and poetic eulogy, to time. Time is typically the force of entropy, the bearer of death, often imagined as an old man wielding a scythe. But in the pataphysical dimension, time collides with space and flows in two directions. As Jarry postulates in his essay on how to construct a time- exploring machine, "Duration is the transformation of a succession into a reversion. In other words: THE BECOMING OF A MEMORY."[14]

In a chapter of *Faustroll* on the subject of heraldry and Tarot cards, during an interview of a Geometer by Mathetes (a disciple of Christ whose name is etymologically akin to mathematics), Chronos, called Father of Time or of Life, takes form from perpendicular vertical and horizontal lines becoming the upper case letter H.[15] This generative spinning of dimensions imitates the motion of the simultaneously plus sign, minus sign, and circle which is also said to be the "egg or the zero" that is the fruit of what Jarry calls the mutual "fecundation" of signs that forms the letter "H". Jarry specifies that this is a "demonstration (…) of the identity of opposites" performed by the gyroscopic *bâton-à-physique*, described in this exegesis of time and space as a "mechanical engine". The *bâton-à-physique*, the stick or staff whose manifestations of "meta" and "pata" equate it to divinity, to Ubu, to the Antechrist, to King Ubu's scepter, as well as to the protean phallus, is thus also an engine designed to facilitate navigation from the mathematical signs of the manifest content to the incongruously poetic subversive semiological signs. According to Jarry's intertextual characters viewing the baton's acrobatics, this shifting, spinning signifier is itself an imaginary solution: at once a line and a circle, negative and positive, man and woman, sex and spirit, and more. Its plurality of content is contained in its geometric motion.[16] Time initially becomes spatialized and approaches the spatial imaginary of the Web with its domains and sites.[17] In his "Commentary to Help in the Practical Construction of the Time-Machine," Jarry plans the construction of a machine that would explore time by means of exploring space. For the Time-Machine, the present is multi-dimensional and appears as a sequence of solid geometric shapes. But, as Jarry predicted, it now, eventually, organizes flowing streams of information as blogs, feeds, video, and especially data, so that we are able to see on our screens what is occurring everywhere at once. He hypothesized that the ideal exploration of this pataphysical space would require that we "remain immobile in absolute Space, along the Flow of Time."[18] He surmised that this coincidence of flow and synchroneity as principles of information sharing allow for varying and sometimes undetermined speeds. Evidence of this relativity underlies *The Supermale* (*Le Surmâle*) in a race by a 5-man cycle team nourished by *Perpetual-Motion-Food* against a locomotive.[19]



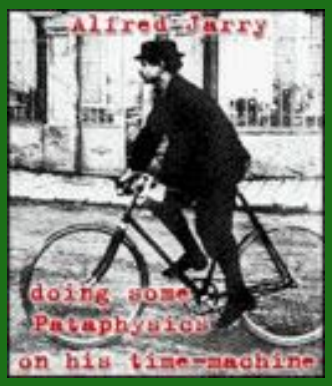
5-Man Cycle Team

The bicycle was an invention of Jarry's time (exhibited at the 1900 Paris World's Fair) and his never-paid-for *Clément Luxe* was his pride and joy. It was built for speed and it allowed him a gateway into the supplementary universe and solutions he imagined. His bike wheels rotation was equivalent to the new time-space physics generated by the *bâton-à-physique*.



Jarry on his Bicycle

His machine became an extension of himself, a cyborgian entity. He often cycled the forty miles between Paris and his shack on the banks of the Seine in Le Coudray. This vehicle data becomes even more pertinent: Jarry used to race the Paris train on a stretch where the track and the road were in close parallel.



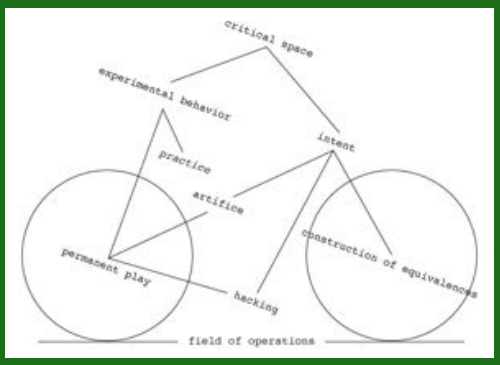
Jarry on his Bicycle Time-Machine

He would probably approve of the diagram of a bicycle composed of two circles and an outline of a house, the accidental representation of his scientific perspectives on watches as timepieces and facades of houses as trapezoids, elucidated in his definition of 'Pataphysics.



Scientific American 1973

Jobs, a self-described " bike lover, a curiosity jockey, and a techno-optimist," adopts a logo combining his belief and his brand.



Bicycle Diagram

Fast-forward to about a century later when Steve Jobs comes across a 1973 Scientific American study on a bicycle providing superhuman velocity to its human rider.[20]



Bicycle and Computer Logo

He takes the data and states, "What a computer is to me is it's the most remarkable tool that we've ever come up with, and it's the equivalent of a bicycle for our minds (…) a metaphor that I (…) want to shake in the face of every false prophet pedaling techno-dystopia." The pataphor already existed in the machinery of Jarry's patadata.



Bicycle and Computer Branding

"*Ainsi que le coquelicot et le pissenlit à la fleur de leur âge sont fauchés par l'impitoyable faux de l'impitoyable faucheur qui fauche impitoyablement leur pitoyable binette,—ainsi le petit Rensky a fait le coquelicot (…).*"

"As the poppy and the dandelion in the flower of their age are scythed by the pitiless scythe of the pitiless scyther who pitilessly scythes their pitiful mugs, so little Rensky has played the role of the poppy (…)."

Ubu Roi Act IV, scene v

Another prototypical semantic search generated by the *Ubu Roi* text sample connects a network linked to the flower *coquelicot*, "poppy," itself already surprisingly linked semantically to the scythe. The poppy continues to be a metaphor for death and for war, used widely today as a reference to the horrors of Flanders Field during World War I.[21] This flower is also a metaphor for all manner of narcotics and directs us to the pataphor of opium smoking and a hallucinatory ether breathing and hashish session in Jarry's novel *Days and Nights, novel of a deserter* (*Les Jours et les nuits, roman d'un déserteur*).[22] In this case, the data are collected from Jarry's own military service, cut short in 1895 at thirteen months. The poppy, chosen by Ubu to give meaning to a fallen soldier's sacrifice, equally stands as a promise of clairvoyance and of sleep. The seemingly delirious comedic raving of King Ubu mirror the serious and tragic hallucinations of soldiers who desert the battlefield in another way. Jarry himself, by means of subterfuge, taking swigs of liver-damaging, skin-yellowing picric acid (a battlefield explosive and an element mentioned in alchemical writings), exited the military ignobly via the infirmary. Additional data informing this sequence are statements in the deathbed letter Jarry wrote to his friend Rachilde, specifying that it was Ubu speaking, thus creating another mirror image of himself. The letter articulates his theory that death is the ultimate act of creativity. In powering himself down, the machine-man, as he saw himself, by means of his decomposing brain gives himself over to sleep and dreams that function beyond death. In this act of "going beyond" to a supplementary universe, Jarry as Ubu announces that he follows a pathway to Paradise and that this process communicates his esthetics.[23] The scarlet color of the poppy also redirects the viewer to a supplementary universe, signifying a promise of resurrection after death, all things being equal.

The relay of semantic information in Ubu's discourse is extended to another flower, *le pissenlit*, the "dandelion." This yellow flower and its red counterpart take us to yet another semantic portal, that of Tarot and of alchemy. Alchemical flow and transformations, death and resurrections: these pataphysical interfaces form networks throughout Jarry's intertext. The *pissenlit*, in French, also links, in apparent contradiction, to the ubuesque or grotesque and slapstick schoolboy humor Jarry favored. Ubu's data, dandelions, become the reader's or viewer's metadata as an emblem of an army of youth cut short, while it self-reflexively and paradoxically leads us beyond metadata to its literal translation, that is coincidentally patadata, of "pissing in the bed," yet another way of collecting information about young boys playing soldier. Such a cross-referencing of lexical meaning with deconstructing language to discover new meaning by the agency of a search engine that is the text itself opens the virtual and imaginary potential of Web semantics.

Both flowers invoked by Commander-in-Chief Ubu, the poppy and the dandelion, are potentially active for further meaning in Jarry's reconfiguration of models of interactivity, connectivity, and interoperability. Indeed, the flowers are cut down *à la fleur de leur âge*, that is, in the prime of life, but literally, in French, "in the flower of their age." All of these relationships build context. Word play, puns, polysemic and often irreverent neologisms, ambiguity, and slightly modified repetitions such as the *impitoyable* set reinforce the deliberately imprecise systems of 'Pataphysics. Furthermore, the "*binette-impitoyable*" semantic field serves as a conduit to the texts of a major figure of French Romanticism, Gérard de Nerval (1808–1855), a generation older than Jarry who read his works. Nerval referred to his absent mother's face as a "*binette*" and wrote that women, especially with Valois bloodlines, are "*impitoyables*".[24] Jarry's fraught relationships to his mother and other women permeate his fiction, including, by deviation, Ubu's superficially farcical discourse. Jarry's books are always appropriating and cross-referencing his other and other's books. In his *César-Antechrist*, for example, Jarry alludes very clearly to Lautréamont's *Chants de Maldoror* when the Templar

addresses the *bâton-à-physique* as "uprooted Phallus" and admonishes it "not to jump around like that".[25]

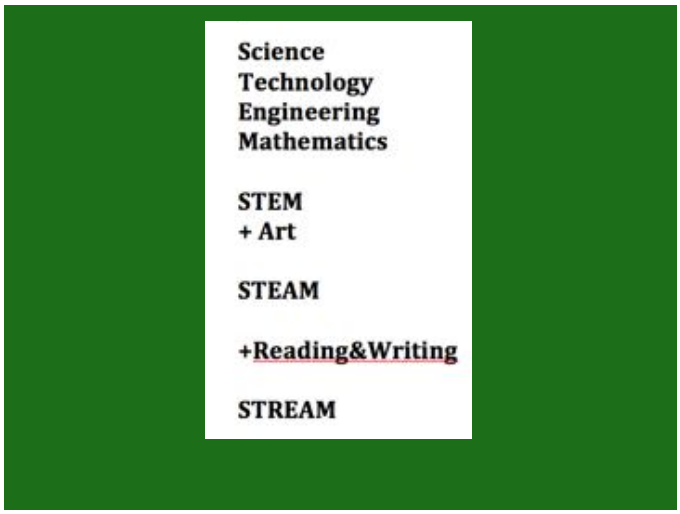
Jarry might have appreciated Wikipedia's "disambiguation pages", which are places where you are not expected to end a search or to even end up.[26] A disambiguation page is a sort of in-between page that lists all the places with the same name where a link may have intended to take you. These are pages for cross-reference and unexpected connections. Similarly, by turning the binoculars around, Jarry studies laws governing exceptions. He mastered the Hapax Legomenon, constituting words appearing only once in an entire language or in an author's entire corpus.[27] Engineering a lexicon of words such as *instintestincts*, *rhizomorphododendron*, and *parallelirésultante*,[28]



Jarry foresaw, in a way, the disambiguation page, perhaps even imagining a website with no links to it on the internet. A critical appropriation of Jarry's own semantic configurations would designate his irony as rapier-sharp. Humor is one of the languages at work in the battlefield enunciation and always already elsewhere in Jarry's writing. Humor is a pataphysical tool for creating disruption of certain meaning. Management of information is sometimes unpredictable, as demonstrated by the experiment in Jarry's 1902 futuristic novel *The Supermale, modern novel* (*Le Surmâle, roman moderne)*. A robot is built in order to inspire love in the superhuman robotic protagonist. Tragi-comically, with Jarry's typically dark humor, the Supermale inspires love in the machine that unexpectedly falls in love with him, reversing the electrical current and electrocuting him.[29] Jarry knew that computers would have to learn in order to reprogram themselves, in order to expand the creativity of algorithms, in order to function by induction.



Jarry also realized that the scientists, technologists, engineers, and mathematicians conspicuously at work in his novel would have to add art to make STEM into STEAM,



and eventually welcome reading and writing to develop STREAM, an image suitable for the flow from the Internet to the Web to the Semantic Web, and applicable to his avant-garde pataphysical procedures and formulas.[30]

This sort of data display transfers data packets between semantic zones, like a bobbin shuttling from data to metadata to patadata. The science of imaginary solutions encompasses the flow and vortex of information also represented by the experiments with water, wave function, electromagnetism and luminiferous ether studied by Jarry's pataphysician extraordinaire, Doctor Faustroll.





Véritable portrait de Monsieur Ubu

The spiral imprinted on Ubu's iconic belly traces the motion of such a swerve of meaning from an original direct denotation of first-degree data to unintended languages. Jarry's repurposing of the ancient Lucretian atomic theory of matter, the *clinamen*, makes an unforeseen deviation into an origin myth for the digital age.[31] Pataphysics is an experimental esthetic and science, and is a prophetic textual and speculative machine that could define the internet, the Web, information technology, computer operations, and the move from sharing documents to sharing data. As used and interpreted by Jarry, the word "pataphysics" itself is a pivotal and radical neologism.[32] "Pataphysics" was Jarry's "neo" or "new" "logos", his new word or new meaning, his new philosophy that governs and develops the universe. The logic does not lead from "because Jarry" or "because Pataphysics", then "Semantic Web". Neither can we hypothesize that "if not Pataphysics" then never "Semantic Web". Rather, Jarry inaugurated a Zeitgeist propitious for the development of specific strands, coincidences, and principles that pioneer today's discoveries and exigencies. His "Elements of Pataphysics"[33] plagiarize by anticipation elements of today's Web science. Ubu and Faustroll are patacessors of our colleagues. Jarry the vanguard poet, essayist, novelist, dramaturge, visual artist, and almanach writer went beyond the codes and meta codes he inherited of Realism, Naturalism, and Symbolism. He went beyond into a non-linear, supplementary, elastic domain that pioneered the way towards Dada, Modernism, Theater of the Absurd, performance art, Zippy the Pinhead comic strips, punk, and the merging of genres. Jarry's vast erudition extended to science of antiquity and to contemporary physics and mathematics as well, and especially the theories and research of Lord Kelvin, William Crookes, Charles Vernon Boys, James Clerk Maxwell, Henri Bergson, and Jules Henri Poincaré.[34]

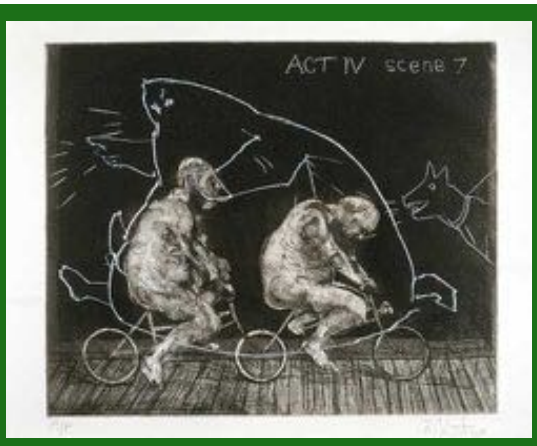
Prescient in so many fields—quantum physics, dark matter, and black holes, multiple colliding universes of quantum mechanics, new dimensions violating cause and effect, non-Euclidean calculations, infinite dimensional space of wave function[35]— Alfred Jarry integrates his fragmented narratives, segmented syntax, building blocks of syllables, and concrete linguistic morphologies into data organization that foregrounds viewers' perspectives, predicting the mirroring of multiple users in front of multiple electronic screens all viewing the same immaterial copied image and uploaded information. In the scientific theories and discoveries of his day, Jarry found common ground with a supplementary universe of interconnected energies described by field theory and electromagnetism. This intersection of esthetic and scientific platforms depends on a system of identity. Identity and equivalence are fundamental to 'Pataphysics as both a technique and a philosophy. Mirroring is a blueprint for Jarry's development of doubling and multiplying language, psychology of characters, fracturing a character into duos and trios and more, and explicitly holding up imaginary mirrors to his readers and audiences. This model remains evident in interpretations by other multimedia artists influenced by Jarry, for example the South African William Kentridge in his 1996 series of etchings "Ubu Tells the Truth", prepared for an exhibition on the centenary of *Ubu Roi*'s 1896 Parisian première."[36]



William Kentridge "Ubu Tells the Truth" (Act V, scene 4)

In Jarry's texts, from his time-exploring machine and his love-inspiring machine to his cosmogonic painting machine, from the locomotive to the strength-testing meter and the multi-man racing cycle, from the debraining machine to his ubuesque merdre-pumping machine to the phonograph that gives orders to the Supermale and is likened to a Siren in a previous work ten years earlier, transactions take place between machines and fictional characters, intuitive of the Semantic Web where computers would process and analyze all the data between computers themselves and the humans who interact with them. These authoritative machines, reproduced in altered forms across Jarry's complete works, simulate mirrored websites working through space and time to deliver content. Charting the occurrence of all these machines parallels the organization of occurrences of the term "pataphysics" and its variants, beginning with a first publication by Jarry in 1893 in his prototypical "Ubu" play *Guignol* ("Puppet Play"), and incorporated the following year into a longer work with further references to

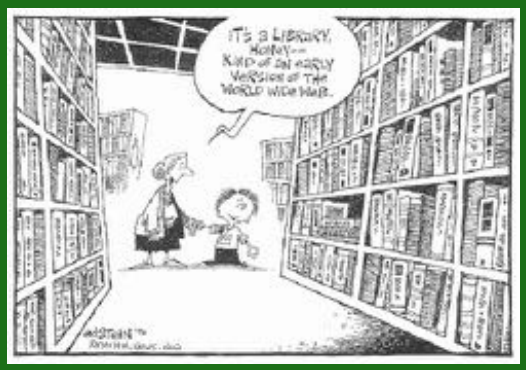
"pataphysics".[37] The machines function and, at times, malfunction. What does the pataphysical machine mean and how does it signify? What, Jarry wonders, is the ontological status and semiotic potential of the machine? Jarry understood the capacities and the storage capabilites of machines. He saw himself as a machine, with his motor giving out at the end.[38] Throughout his career and his opus, he used machines in texts and texts as machines[39] to be transmitted as data by a machine of mirrors, just as he used books and their storage capacity as data to imagine an internet library not unlike the massive database of books scanned by Google,[40] a web of copies, a literary map, a cultural geography of links.



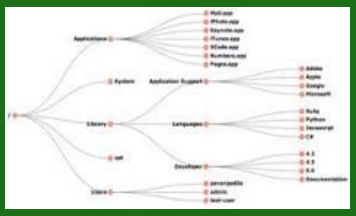
William Kentridge "Ubu Tells the Truth" (Act IV, scene 7)

In *The Supermale*, the entire novel serves as a library of all books. It is a text assembled from quotations and references that bear witness to Jarry's prodigious knowledge. This compilation of erudite citations referring to a litany of men with superhuman strength, from antiquity to the modern age, propels the narrative action and mirrors the story itself. Pertinently, in *Faustroll*, stored information becomes the list of twenty-seven so-called equivalent books (*les livres pairs*) in the Doctor's library.[41] All of the seemingly incongruous and even incompatible books exist equally and concurrently in Jarry's coherent mind, like the synapses of a brain. The listed inventory of books serves as data, the chapters of the novel devoted to an exploration of the books serve as metadata, and the voyage around the archipelago of books become islands serves as the patadata of the epic pataphysical cruise within the complex system that is the book.

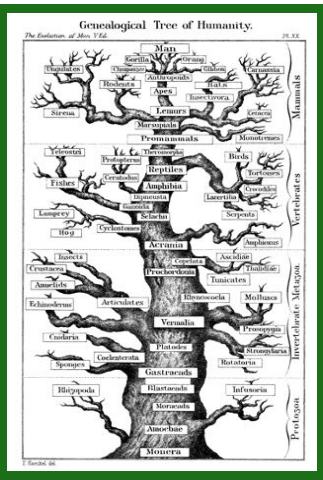
As the itinerary in the archipelago verifies, the design Jarry intuits is no longer the hierarchical "tree" format used for centuries, with a trunk and branches, such as one used to depict a "family tree".[42]



Perceptive and canny, Jarry intuits the diagram of a network, of a web, of a rhizomatic organism. Disparate and even contradictory components join together. 'Pataphysics manifests itself via a network of reappearing phonemes, morphemes, syllables, and hybrid words; of mathematical data visualized in fictional narrative; of repeating and reappearing textual passages and of unfinished endings.

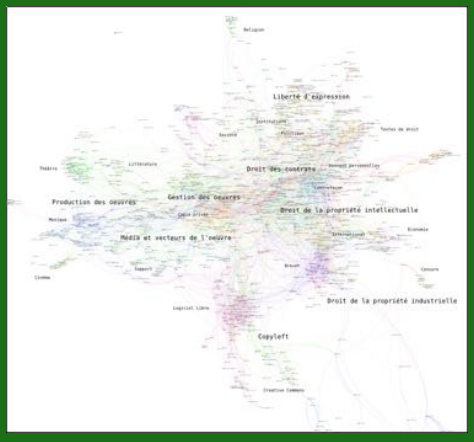


Tree Diagram # 1

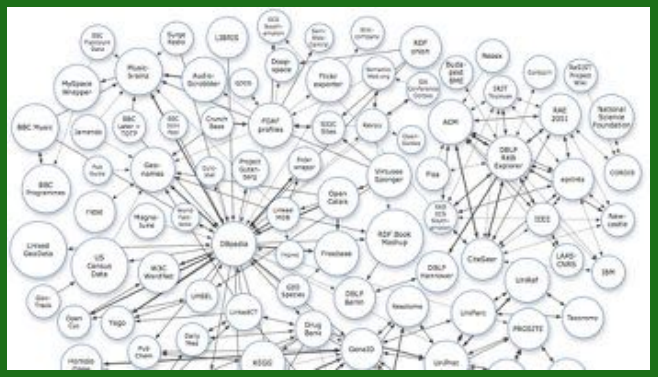


Tree Diagram # 2

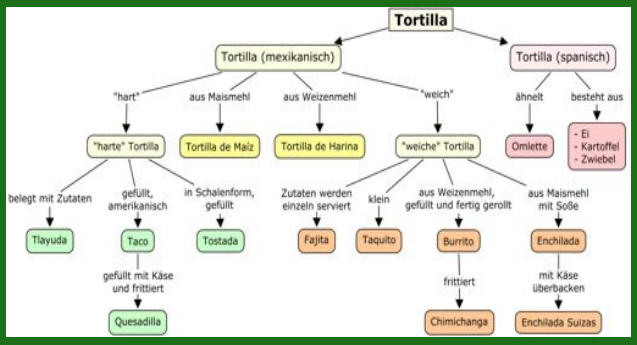
His texts are condensed and elegant intertexts. His diagrammatic subtexts are decentralized, asymmetrical, and non-hierarchical. Jarry's web of meaning foretells internet program files and information sharing. Pataphysician par excellence, Faustroll visits an archipelago, connecting the scattered, clustered dots as he goes along. The archipelago functions as a geospatial virtual environment, with the space between islands indicating a semantic distance. Each visit to an island in the archipelago provides entry into a supplementary literary or artistic universe based on the information in his library.[43]



Semantic Web Diagram #1



Semantic Web Diagram #2



Semantic Web Diagram #3

Jarry's faustrollian dinghy sets off "surfing" heterogeneous fields and domains of language, of information, of appropriation, of copies, surfing a multidimensional imaginary archipelago like points to be connected in a design network and web structure of format and links. Jarry was a rule-breaker and a creator of new rules. Doctor Faustroll and his crew travel from island to island in a sieve, a scientific exploration, as well, of the surface tension of water, surfaces of no curvature, and other experiments. The vessel itself is a testimony to the validity of paradox. Moreover, we learn that the odyssey to exotic islands takes place on dry land and that it begins and ends in Paris. Jarry once again sets universes colliding, data confronting patadata. Is it dry or wet, land or sea, will the boat sink or float? Before moving to Paris for highschool, the child Alfred Jarry lived in Brittany where his mother had copies of *Le Magasin Pittoresque* in which Jarry discovered many sources of inspiration for his writing and in which the curious boy would have seen wood-engraved or lithographed images of the Paris floods of 1876 and 1883, each over six meters of inundation in the capital.[44] Flooded Parisian streets and sidewalks would become the setting for an imaginary cruise where opposites are equal and contradictions are both true.[45]



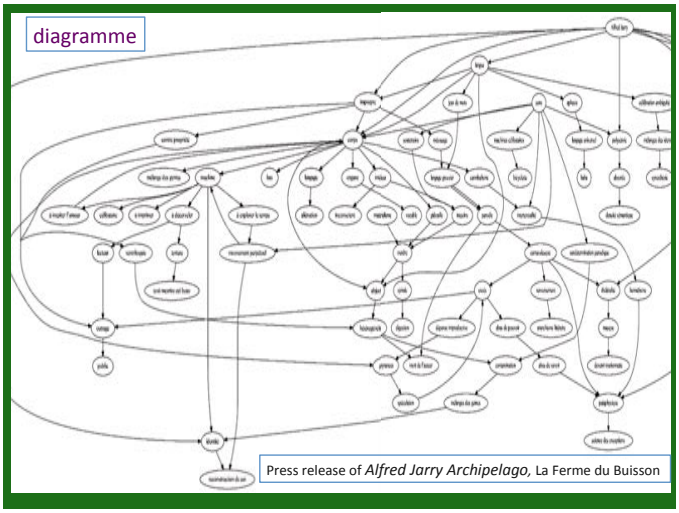
1883: A view of the flooded street outside the Parisian workshop as construction begins for the bequeathing ceremony

His exploration of new ontologies was logically messy and at the same time precise. What a coincidence that Web Ontology Language, a Semantic Web language whose documents are known as ontologies, is abbreviated OWL:[46] ironically and serendipitously, the *clinamen* is at work. Jarry cohabitated with live owls; they were his mascots, his trusty sidekicks. The owl, *le Hibou*, of course also inhabits Jarry's fiction.[47] It is no surprise that someone eventually wrote a play called *Hibou Roi*.[48]



The hilarious adventures of the heir of Alfred Jarry

The nocturnal owl possesses night vision, highly valued by Jarry's characters, and since antiquity it has symbolized not only metamorphosing but apparently contradictory attributes. Ubu and the owl embody and exemplify the elliptical curves specified as representing the real world in Jarry's definition of Pataphysics. They resemble wine barrels. Ubu describes himself as the "isomorphe" of the barrel in all its beauty.[49] Jarry's navigation of data, metadata and patadata demonstrates at all times his pataphysical definition of "simplicity" equaling "condensed complexity".[50] Even on his deathbed, Jarry continued to build vocabularies and link data. In the end, his legendary final request for a toothpick, a *cure-dent*, engages a semantic selection of the scythe, the dragoon, the saber, the military, death, war, and all of the interrelated searches they target. In military parlance, a *cure-dent* is a "bayonet". His context-understanding irony establishes a talisman of sorts offering passage via a toothpick to the supplementary universe and imaginary solutions. And Alfred Jarry, when he set sail into "ethernity"[51] at age 34, broke and cut down/"*fauché*" in the prime of his life/"*à la fleur de son âge*", as he predicted, found a paradisaical solution by means of his decomposing brain, that ultimate network of knowledge and brainpower.



Press release of *Alfred Jarry Archipelago,* La Ferme du Buisson



Brain neurons



The rules and algorithms of 'Pataphysics—deviation, conjoining of opposites, and paradox – are semantic tools to harness potential, to navigate and integrate the flow of information, of data, of images, to build interconnections and frames of reference. Jarry's formulation of 'Pataphysics at the critical juncture between the nineteenth and the twentieth centuries resonated with the potential momentum of deconstructing time and space, much as Paris was newly opened to wide vistas, efficient thoroughfares, accessible and uncluttered corridors of entry and exit; and Parisians attending the 1889 and 1900 World Fairs were delighting in novel, cutting-edge, state-of-the-art machinery and technology.[52] This decisive paradigm shift disrupts and challenges our understanding and depiction of knowledge and its complexities.

REFERENCES

[1] [1] Stillman, Linda Klieger. *Alfred Jarry*. Boston, Massachusetts: Twayne Publishers, 1983, pp 124-138.

[2] [2] Jarry, Alfred. *Gestes et opinions du docteur Faustroll, pataphysicien, roman néo scientifique*. *Oeuvres completes*, t. I, textes établis, présentés et annotés par Michel Arrivé. Paris: Gallimard, Bibliothèque de la Pléiade, 1972.

[3] [3] Jarry, Alfred. Definition of 'Pataphysics, from *Exploits and Opinions of Doctor Faustroll, Pataphysician*, chap. 8, in *Evergreen Review Reader 1957-1967: A Ten-Year Anthology* (New York: Grove Press, Inc, 1968), pp. 310-311. Original French quotation:

[4] "Un épiphénomène est ce qui se surajoute à un phénomène. La pataphysique dont l'étymologie doit s'écrire et l'orthographe réelle 'pataphysique, précédé d'une apostrophe, afin d'éviter un facile calembour, est la science de ce qui se surajoute à la métaphysique, soit en elle-même, soit hors d'elle-même, s'étendant aussi loin au-delà de celle-ci que celle-ci au-delà de la physique. Et l'épiphénomène étant souvent l'accident, la pataphysique sera surtout la science du particulier, quoiqu'on dise qu'il n'y a de science que du général. Elle étudiera les lois qui régissent les exceptions et expliquera l'univers supplémentaire à celui-ci; ou moins ambitieusement décrira un univers que l'on peut voir et que peut-être l'on doit voir à la place du traditionnel, les lois que l'on a cru découvrir de l'univers traditionnel étant des corrélations d'exceptions aussi, quoique plus fréquentes, en tous cas de faits accidentels qui, se réduisant à des exceptions peu exceptionnelles, n'ont même pas l'attrait . de la singularité. DÉFINITON: La pataphysique est la science des solutions imaginaires, qui accorde symboliquement aux linéaments les propriétés des objets décrits par leur virtualité. La science actuelle se fonde sur le principe de l'induction: la plupart des hommes ont vu le plus souvent tel phénomène précéder ou suivre tel autre, et en concluent qu'il en sera toujours ainsi. D'abord ceci n'est exact que le plus souvent, dépend d'un point de vue, et est codifié selon la commodité, et encore ! Au lieu d'énoncer la loi de la chute des corps vers un centre, que ne préfère-t-on celle de l'ascension du vide vers une périphérie, le vide étant pris pour unité de non-densité, hypothèse beaucoup moins arbitraire que le choix de l'unité concrète de densité positive eau ? Car ce corps même est un postulat et un point de vue des sens de la foule, et, pour que sinon sa nature au moins ses qualités ne varient pas trop, il est nécessaire de postuler que la taille des hommes restera toujours sensiblement constante et mutuellement égale. Le consentement universel est déjà un préjugé bien miraculeux et incompréhensible. Pourquoi chacun affirme-t-il que la forme d'une montre est ronde, ce qui est manifestement faux, puisqu'on lui voit de profil une figure rectangulaire étroite, elliptique de trois quarts, et pourquoi diable n'a-t-on noté sa forme qu'au moment où l'on regarde l'heure ? Peut-être sous le prétexte de l'utile. Mais le même enfant, qui dessine la montre ronde, dessine aussi la maison carrée, selon la façade, et cela évidemment sans aucune raison; car il est rare, sinon dans la campagne, qu'il voie un édifice isolé, et dans une rue même les façades apparaissent selon des trapèzes très obliques. Il faut donc bien nécessairement admettre que la foule (…) est trop grossière pour comprendre les figures elliptiques, et que ses membres s'accordent dans le consentement dit universel parce qu'ils ne perçoivent que les courbes à un seul foyer, étant plus facile de coïncider en un point qu'en deux. Ils communiquent et s'équilibrent par le bord de leurs ventres, tangentiellement. Or, même la foule a appris que l'univers vrai était fait d'ellipses, et les bourgeois mêmes conservent leur vin dans des tonneaux et non des cylindres." *Oeuvres complètes*, t. I, pp. 668-669.

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[6] [4] An unusually extended metaphor that creates its own context, invented by Pablo Lopez (Paul Avion) based on Jarry's science of 'pataphysics. http://www.pataphor.com/whatisapataphor.html#top

[7] [5] https://tw.rpi.edu/weblog/2008/05/28/patadata/

[8] [6] James Hendler and Andrew Hugill, http://sciencespot.co.uk/tabling-a-new-concept-in-creative-search.html

[9] [7] Jarry, Alfred. *Ubu Roi, Oeuvres complètes*, t. I, pp. 345-398.

[10] [8] *Ubu Roi* (Act IV, scene v), *Oeuvres complètes*, t. I, p. 386.

[11] [9] Jarry, Alfred. *Ubu Roi*. Translated by Barbara Wright. USA: New directions Book, 1961, pp 118-119. (Originally published London: Gaberbochus Press, 1951).

[12] [10] Stillman, Linda. *Modern Narrative Technique: Jarry, the pre-text*, *Sub-Stance*, Vol. II, No. 36, (1982), pp. 71-81.

[13] [11] *La Dragonne, roman. Oeuvres complètes*, t. II, Édition établie par Henri Bordillon avec la collaboration de Patrick Besnier et Bernard Le Doze, pp. 419-524. This is Jarry's final, and unfinished, work. On May 27, 1906, he dictated a detailed outline to his sister Charlotte who completed the text.

[14] [12] "Il est d'usage d'appeler MONSTRE l'accord inaccoutumé d'éléments dissonants ; le Centaure, la Chimère se définissent ainsi pour qui ne comprend. J'appelle monstre toute originale inépuisable beauté." / "It is conventional to call 'monster' any blending of dissonant elements. I call 'monster' every original inexhaustible beauty." *L'Ymagier*, *Oeuvres completes*, t. I, p. 972.

[15] [13] "La SCIENCE avec une grande SCIE…" *Le Surmâle*, Chapitre ix, *Oeuvres complètes*, t. II, p. 247. While the final sentence of *Faustroll* is "*La Pataphysique est la science*…" (Jarry's emphasis), a few years later, in *Le Surmâle*, he interconnects his science with the phallic instrument.

[16] [14] "Commentaire pour servir à la contruction pratique de la machine à explorer le temps"/"How to Construct a Time Machine", *Oeuvres complètes*, t. I, pp. 735-743. "La durée est la transformation d'une successon en une réversion. C'est-à-dire: LE DEVENIR D'UNE MÉMOIRE" (Jarry's emphasis). *Chapitre* v, p. 743. H. G. Wells's science fiction novel *The Time Machine* appeared in a French translation by Henri-D Davray in the *Mercure de France* in 1898-99. Jarry's text

appeared in the (with Faustroll's signature) in February, 1899. *Mercure de France*, No. 110, pp. 387-396.

[17] [15] *Oeuvres complètes*, t. I, pp. 729-730.

[18] [16] Stillman. Alfred Jarry. Chapter Two, "Pataphysically Speaking…," pp. 16-42.

[19] [17] http://www.newyorker.com/magazine/2014/10/20/crooner-rights-spat

[20] [18] "Commentaire pour servir à la contruction pratique de la machine à explorer le temps"/"How to Construct a Time Machine", *Oeuvres complètes*, t. I, p. 736.

[21] [19] "La course des dix mille milles," *Le Surmâle*, *chapitre* v, *Oeuvres complètes*, t. II, pp. 217-232.

[22] [20] http://www.bikeboom.info/efficiency/

[23] http://www.cyclelove.net/2013/03/steve-jobs-on-why-computers-are-like-a-bicycle-for-the-mind/

[24]

[25]

[26] [21] https://en.wikipedia.org/wiki/In_Flanders_Fields

[27] [22] "Les Propos des Assassins,"Les Jours et les nuits, Livre V, chapitre iv, Oeuvres complètes, t. I, pp. 821- 829.

[28] [23] Il croit que le cerveau, dans la décomposition, fonctionne au delà de la mort et que ce sont ses rêves qui sont le Paradis. Jarry, *Oeuvres complètes*, t. III, Édition établie pas Henri Bordillon avec la collaboration de Patrick Besnier et Bernard Le Doze et la participation de Michel Arrivé, pp. 616-617. (Les lettres ont été publiées par Rachilde elle-même, dans *Alfred Jarry ou le Surmâle de lettres*, 1928).

[29] [24] Joseph Citrouillard. *Les Binettes contemporaines*, Tome II, Paris: Imprimerie Walder, 1854. Also, **http://laporteouverte.me/2016/03/19/les-binettes-contemporaines-gerard-de-nerval/.**

[30] [25] "Le Templier: Phallus déraciné, NE FAIS PAS DE PAREILS BONDS!" (Jarry's emphasis). *Oeuvres complètes*, t. I, p. 289.

[31] [26] http://en.wikipedia.org/wiki/Wikipedia:Disambiguation

[32] [27] http://en.wikipedia.org/wiki/Hapax_legomenon

[33] [28] "Les propos des Assassins," *Les Jours et les nuits*, *Oeuvres complètes*, t. I, pp. 827-828.

[34] [29] "La Machine amoureuse," *Le Surmâle*, chapitre xiv, *Oeuvres complètes*, t. II, pp.265-271.

[35] [30] "Our desire is to connect the worlds of leading scientists with international artists through 'creative collisions', encouraging both fields to inspire and challenge each other, and pushing the boundaries of their traditional roles and methodologies," Arts@CERN's director: the European Center for Nuclear Research. March 2016, http://www.smithsonianmag.com/smart-news/cern-seeks-international-artists-full-time-residency-180958448/?no-ist

[36] [31] Jarry defines the *clinamen* as *la bête imprévue*, ("the unforeseen beast"), that causes a Painting Machine in the Palace of Machines to ejaculate paint on the walls in a modern deluge. So doing, the Painting Machine creates a supplementary universe. "Clinamen," *Faustroll*, chapitre xxxiv, *Oeuvres complètes*, t. I, p. 714. The Lucretian theory of atoms posits: "When atoms move straight down through the void by their own weight, they deflect a bit in space at a quite uncertain time and in uncertain places, just enough that you could say that their motion has changed. But if they were not in the habit of swerving, they would all fall straight down through the depths of the void, like drops of rain, and no collision would occur, nor would any blow be produced among the atoms. In that case, nature would never have produced anything." https://en.wikipedia.org/wiki/Clinamen

[37] [32] "Pataphysics" first appeared in print in a marionette play staged by Jarry and his highschool classmates in 1889.

[38] [33] "*Éléments de Pataphysique*," Title of Book II of *Faustroll*. *Oeuvres complètes*, t. I, pp. 668-673. This chapter (Book II, chapter viii) contains the definition of pataphysics (cited above in note 3).

[39] [34] Stillman, Linda Klieger. Physics and Pataphysics: the Sources of *Faustroll*. *Kentucky Romance Quarterly*, Vol XXVI, No 1, 1979, pp. 81-92.

[40] [35] Stillman, Linda Klieger. *Alfred Jarry*. Boston, Massachusetts: Twayne Publishers, 1983, p. 38.

[41] [36] Portfolio of eight etchings (1996-97). Hardground, softground, aquatint, drypoint and engraving, 14 x 19 1/2 inches. Kentridge, also studied Jarry's contemporary, Poincaré. Both artists were particulary intrigued by the mathematician and physicist's theories about relativity, time and clocks.

[42] [37] From his early writings to his final works, Jarry reformulates and reapplies pataphysical axioms.

[43] "Pataphysics" or "pataphysician" or another variant occurs, most notably, in *Guignol* (1893), *Les Minutes de sable mémorial* (1894), *César-Antechrist* (1895), *Ubu roi* (1896), *Ubu cocu* (1898, published posthumously 1944), *Gestes et opinions du docteur Faustroll* (1898, published posthumously 1911), *Les Jours et les Nuits* (1897), *L'Amour en visites*, "*Chez Madame Ubu*"(1898), *Ubu enchaîné* (1900).

[44] [38] "Il va s'arrêter tout doucement comme un moteur fourbu." Deathbed letter to Rachilde, May 28, 1906.

[45] [39] Stillman, Linda Klieger. Guest Editor: "Machines in Texts, Texts as Machines, " *l'esprit créateur*, Volume XXVI, No 4, Winter 1986, Introduction, pp. 3-5.

[46] [40] http://www.newyorker.com/magazine/2007/02/05/googles-moon-shot

[47] [41] The list of Faustroll's books:

[48]

[49] BAUDELAIRE, un tome d'EDGARD POE, traduction.

[50] BERGERAC, Œuvres, tome II, contenant l'Histoire des États et Empires du Soleil, et l'Histoire des Oiseaux.

[51] L'*Évangile* de SAINT LUC, en grec.

[52] BLOY, Le Mendiant ingrat. [Le Désespéré]

[53] COLERIDGE, The Rime of the ancient Mariner.

[54] DARIEN, Le Voleur. [Biribi]

[55] DESBORDES-VALMORE, Le Serment des petits hommes.

[56] ELSKAMP, Enluminures. [Salutations dont d'angéliques]

[57] Un volume dépareillé du *Théâtre* de FLORIAN.

[58] Un volume dépareillé des *Mille et Une Nuits*, traduction GALLAND.

[59] GRABBE, Scherz, Satire, Ironie und tiefere Bedeutung, comédie en trois actes.

[60] KAHN, Le Conte de l'or et du Silence. [Le Livre d'images]

[61] LAUTRÉAMONT, Les Chants de Maldoror.

[62] MAETERLINCK, Aglavaine et Sélysette. [Pelléas et Mélisande]

[63] MALLARMÉ, Vers et prose. [Divagations]

[64] MENDÈS, *Gog*.

[65] L'*Odyssée*, édition Teubner.

[66] PÉLADAN, *Babylone*.

[67] RABELAIS.

[68] JEAN DE CHILRA, L'Heure sexuelle. [La Princesse des ténèbres]

[69] HENRI DE RÉGNIER, *La Canne de jaspe*.

[70] RIMBAUD, Les Illuminations.

[71] SCHWOB, La Croisade des enfants.

[72] Ubu Roi.

[73] VERLAINE, *Sagesse*.

[74] VERHAEREN, Les Campagnes hallucinées.

[75] VERNE, Le Voyage au centre de la Terre.

[76]

[77] In addition, there are artworks: Three engravings; a poster by Toulouse-Lautrec of *Jane Avril*; another by Bonnard; a portrait of Faustroll by Aubrey Beardsley; and an old image, *saint Cado*.

[78] "Plus trois gravures pendues à la muraille, une affiche de TOULOUSE-LAUTREC, *Jane Avril* ; une de BONNARD, *La Revue blanche* [*L'Estampe et l'Affiche*, corrigé en *Peintres Graveurs*] ; un portrait du sieur Faustroll, par AUBREY BEARDSLEY, et une vieille image, laquelle nous a paru sans valeur, *saint Cado*, de l'imprimerie Oberthür de Rennes."

[79] Oeuvres complètes, t. I, p. 661.

[80] [42] Important research has been done on tree and network diagrams by Manuel Lima in his work on information visualization. http://www.mslima.com/myhome.cfm

[81]

[82] [43] There is a diagram of a Pataphysics archipelago in the press release of *Alfred Jarry Archipelago* by La Ferme du Buisson, a national theater and contemporary art center in Marne-la-vallée, France. http://www.lafermedubuisson.com/IMG/pdf/la-ferme-du-buisson_alfred-jarry-archipelago_dossier-de-presse.pdf (pp. 4-5).

[83] [44] Julien Schuh, "*Jarry et Le Magasin Pittoresque*", *L'Étoile-Absinthe,* tournées 123-124. Paris: SAAJ & Du Lérot éditeur, pp. 102-137.

[84] [45] Jarry's inspiration extended to mythology as well. Faustroll's archipelago includes, for example, *L'Île de Bran* (*Faustroll*, Book III, Chapter xii, *Oeuvres complètes*, t. I, pp. 675-677.) In this chapter, Faustroll cites his "*Éléments de pataphysique*" and refers to his sea-faring company as "pataphysicians". The port-of-call is an allusion to an Irish tale of an odyssey by King Bran to an imaginary universe, an "Otherworld" where a sea is perceived as a chariot-ready "flowery plain" by a Celtic sea-god. Bran embarks with his three foster brothers each of whom brings nine men aboard their boat: are Bran's twenty-seven fellow-travelers the source of Jarry attributing twenty-seven books to Faustroll's library? The Otherworld's dry sea corresponds as well to the reference in *Faustroll*, Book IV, chapter xxv to the "terrestrial tide" (*la marée terrestre*).

[85] [46] https://www.w3.org/2001/sw/wiki/OWL

[86] [47] In *Haldernabou*, part of *Les Minutes de sable mémorial*, Haldern asks the owl to protect him, addressing it as "*Hibou ocellé*". *Oeuvres complètes*, t. I, p. 225. Identifying the owl with the ocelet renders it a spirit animal, a totem.

[87] [48] *Hibou Roi* by Fabrice Sigwalt and Flobert appeared in 1976 in the monthly *Scop Magazine*, published by les Éditions de la Grille. https://books.google.com/books?id=EhRdAgAAQBAJ&pg=PA5&lpg=PA5&dq=alfred+jarry+le+hibou&source=bl&ots=cLEUb0bRaA&sig=CVHZT7fg5TuHrPAw_N35ZJFeyrE&hl=en&sa=X&ved=0ahUKEwjWu8DwwOnLAhUM7yYKHYaqBsYQ6AEITTAH#v=onepage&q=alfred%20jarry%20le%20hibou&f=false

[88] [49] L'Art et la Science, in Les Minutes de sable mémorial, Oeuvres complètes, t. I, p. 188.

[89] [50] *Linteau*, *Les Minutes de sable mémorial*, *Oeuvres complètes*, t. I, p. 172. "La simplicité n'a pas besoin d'être simple, mais du complexe resserré et synthétisé (cf. *Pataph*.)"

[90] [51] The neologism "*éthernité*", homophonous with "eternity"/"*éternité*" in French, but merging it with the word "ether", a drug favored by Jarry, is the title of Book VIII, the final chapter, of *Faustroll*. This chapter recounts Doctor Faustroll's death, at the same age at which he was born, his telepathic letters to Lord Kelvin on the subjects of time, space, death, and luminiferous ether, as well as his pataphysical calculations on God's surface measurements. *Oeuvres complètes*, t. I, pp. 724-734.

[91] [52] The 1990 Paris World's Fair (*Exposition Universelle*) featured a Palace of Machines repurposed from the 1889 World's Fair (where many of Thomas Edison's inventions were exhibited), but also the innovations of X-rays, wireless telegraphy, bicycles, automobiles, turbines, and cinema, escalators, and diesel engines. (Jarry was living in Paris and in 1899 had published his *Commentary on How to Construct a Time-Machine*.) The pièce de résistance was the Palace of Electricity (Palais de l'Électrique) connecting the energy flowing to all the pavillions and lighting up the fairgrounds. http://ic.galegroup.com/ic/whic/ReferenceDetailsPage/DocumentToolsPortletWindow?displayGroupName=Reference&jsid=7c80cac443c70bc64f049166655025d5&action=2&catId=&documentId=GALE|CX3446900895&u=mlin_s_martha&zid=6d28932e81a63a4d3eb14c7fe71291fb